
p-adic Delsarte-Goethals-Seidel-Kabatianskii-Levenshtein-Pfender Bound

K. Mahesh Krishna

School of Mathematics and Natural Sciences
 Chanakya University Global Campus
 NH-648, Haraluru Village
 Devanahalli Taluk, Bengaluru Rural District
 Karnataka State, 562 110, India
 Email: kmaheshak@gmail.com
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Abstract: We introduce the notion of p-adic spherical codes (in particular, p-adic kissing number problem). We show that the one-line proof for a variant of the Delsarte-Goethals-Seidel-Kabatianskii-Levenshtein upper bound for spherical codes, obtained by Pfender [*J. Combin. Theory Ser. A*, 2007], extends to p-adic Hilbert spaces.

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1. INTRODUCTION

Let $d \in \mathbb{N}$ and $\theta \in [0, 2\pi)$. A set $\{\tau_j\}_{j=1}^n$ of unit vectors in $\mathbb{R}^d$ is said to be (d, n, θ) -**spherical code** [42] in $\mathbb{R}^d$ if

$$(1) \quad \langle \tau_j, \tau_k \rangle \leq \cos \theta, \quad \forall 1 \leq j, k \leq n, j \neq k.$$

Since

$$\langle \tau, \omega \rangle = \frac{2 - \|\tau - \omega\|^2}{2}, \quad \forall \tau, \omega \in \mathbb{R}^d,$$

we can rewrite Inequality (1) as

$$\|\tau_j - \tau_k\| \geq \sqrt{2(1 - \cos \theta)}, \quad \forall 1 \leq j, k \leq n, j \neq k.$$

Fundamental problem associated with spherical codes is the following.

Problem 1.1. *Given d and θ , what is the maximum n such that there exists a (d, n, θ) -spherical code $\{\tau_j\}_{j=1}^n$ in $\mathbb{R}^d$?*

The case $\theta = \pi/3$ is known as the famous **(Newton-Gregory) kissing number problem**. With extensive efforts from many mathematicians, it is still not completely resolved in every dimension (but resolved in dimensions $d = 1$ ($n = 2$), $d = 2$ ($n = 6$), $d = 3$ ($n = 12$), $d = 4$ ($n = 24$), $d = 8$ ($n = 240$), $d = 24$ ($n = 196560$)) [1, 2, 7, 13, 14, 22, 23, 25–31, 33–37, 40]. We refer [3–6, 8–12, 15, 17, 18, 21, 32, 38, 39, 41] for more on spherical codes. Problem 1.1 has connection even with sphere packing [16]. Most used method for obtaining upper bounds on spherical codes is the Delsarte-Goethals-Seidel-Kabatianskii-Levenshtein